SRM Institute of Science and Technology
Department of Mathematics
21MAB204T-Probability and Queueing Theory
Unit – I
Tutorial Sheet – 1

S.No.	Part – B (Each question carries 8 marks)
1	If the RV X takes the values 1, 2, 3 & 4 such that $2P(X=1)=3P(X=2)=P(X=3)=5P(X=4)$, find the Probability Distribution Function and Cumulative Distribution Function of X. Also find $P(X>2)$.
2	The Probability Distribution Function of a Random Variable X is given below: x: 1, 2, 3, 4 $p(x)$: $4k$, k, $2k$, $3k$ Find (i) k (ii) $P(\frac{1}{2}<X<\frac{5}{2}/X>1)$ (iii) $P(X\le 3)$ (iv) $P(1\le X<2)$
3	The Probability mass function of a discrete RV X is given by $$p(x)=\begin{cases}\frac{1}{4} & or\ x=-2\\ \frac{1}{4} & for\ x=0\\ \frac{1}{2} & for\ x=5\\ 0, & otherwise\end{cases}$$ Find (i) $P(\lvert X\rvert\ge 2)$ (ii) $P(0\le X\le 10)$ (iii) $P(X\le 0)$ (iv) Cumulative Distribution Function
4	The CDF of a discrete RV X is given by $$F(x)=\begin{cases}0 & if\ x<-2\\ \frac{1}{6} & if\ -2\le x<-1\\ \frac{1}{3} & if\ -1\le x<0\\ \frac{2}{3} & if\ 0\le x<1\\ \frac{5}{6} & if\ 1\le x<2\\ 1 & if\ x\ge 2\end{cases}$$ Find (i) the Probability Distribution Function of X. (ii) $P(x\ge -1)$
5	If the CDF of a Random Variable X is $$F(x)=\begin{cases}0 & ,\ x\le 0\\ 2x^2-x^3 & ,0<x<1\\ 1 & ,\ x\ge 1\end{cases}$$ find (i) the PDF of X (ii) $P(\frac{1}{2}<x<\frac{2}{3})$
	Part – C (Each question carries 15 marks)
6	The Probability Mass Function of a Random Variable X is given below: x: 0, 1, 2 $p(x)$: $3C^2$, $4C-10C^2$, $5C-1$ Find (i) the value of C (ii) distribution function of X (iii) the largest value of x such that F(X) < $\frac{1}{2}$ (iv) Find $P(0<X<2\ /\ X>0)$

7	A random variable X has the following PDF, $$f(x) = \begin{cases} 0 & , x < 1 \\ k(x-1) & , 1 \le x \le 2 \\ k(3-x) & , 2 \le x \le 3 \\ 0 & , x > 3 \end{cases}$$ Find (i) the value of k (ii) CDF of X (iii) $P\left(1 < x < \frac{5}{2}\right)$ (iv) $P\ (\lvert X \rvert < 2)$
8	A continuous R.V X that can assume any value between x=2 and x=5 has the density function given by f(x) = k(1+x). Find P (x< 4) and P (3< x <4)